

Sumukesh Reddy Mopuram

B.Tech (CSE), Indian Institute of Information Technology, Sri City

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Professional Summary

Software Engineering undergraduate with strong foundations in **Data Structures, Backend Systems, and Deep Learning**. Experienced in building scalable full-stack applications and RESTful services using **Node.js, Spring Boot, and AWS**. Currently working on transformer-based models for hyperspectral image classification and semi-supervised learning, combining software engineering with applied machine learning research.

Education

- **Indian Institute of Information Technology, Sri City** 2023 – Present
B.Tech in Computer Science and Engineering CGPA: 8.16/10

Experience

- **Software Engineering Intern – Infosys Springboard** Feb 2026 – Apr 2026
Spring Boot, REST APIs, MySQL, JWT
 - Building an end-to-end **Web-Based Coffee Ordering Platform** with multi-role workflows.
 - Designed and implemented **RESTful APIs** for bookings, orders, and payments.
 - Integrated **Spring Security (JWT)**, email verification, and profile completion rules.
 - Designed relational database schema with focus on data integrity and scalability.

Projects

- **Prioritized Semi-Supervised Deep Embedded Clustering** Dec 2025 – Present
Deep Learning, Semi-supervised Learning, Clustering, PyTorch
 - Implementing a semi-supervised clustering model using **prioritized pairwise constraints** to improve latent representation learning.
 - Applied constraint weighting using distance-based priority functions to improve clustering quality.
 - Experimenting with improved loss weighting strategies and optimization techniques for better representation learning.
- **Hyperspectral Image Classification using GSC-ViT** [GitHub] Jan 2026 – Apr 2026
Tensor, Vision Transformers, Deep Learning, Remote Sensing
 - Implementing a lightweight **Groupwise Separable Convolutional Vision Transformer (GSC-ViT)** for hyperspectral image classification using Chandrayaan-2 IIRS lunar dataset.
 - Designed spectral-spatial learning pipeline combining **groupwise separable convolutions and transformer attention**.
 - Implementing efficient training with spectral normalization, patch extraction, and model optimization.
- **Cloud Monitoring Pipeline** Sep 2025 – Dec 2025
AWS Lambda, CloudWatch, Python
 - Built a serverless log monitoring and alerting system on AWS.
 - Implemented automated log analysis and event-driven alerting using AWS Lambda and CloudWatch.
 - Designed scalable monitoring workflows with low operational overhead.

Skills

- **Languages:** C, Java, Python, JavaScript
- **Backend & Web:** Node.js, Express.js, React, REST APIs, Spring Boot
- **Deep Learning & AI:** PyTorch, CNNs, Vision Transformers, RNN, LSTM, Transformers, NLP
- **Machine Learning:** Semi-supervised Learning, Clustering, Hyperspectral Image Processing
- **Databases:** MongoDB, MySQL
- **Cloud & Tools:** AWS (EC2, Lambda, CloudWatch), Docker, Git, GitHub
- **Core CS:** Data Structures and Algorithms, OOP, Operating Systems, DBMS

Positions of Responsibility

- **Contest Manager – Gradient (Programming Club, IIITS)** 2025 – Present
Planned and managed competitive programming contests, ensuring smooth execution and participant engagement.

Strengths

- Solved 700+ Data Structures and Algorithms problems on LeetCode and other platforms
- Strong debugging skills and ability to work across unfamiliar systems
- Experience implementing research papers and experimental deep learning models